FairCKD: Fairness Audit Report

Investigating Bias and Fairness in Machine Learning Models for Automated CKD Diagnosis
M.Tech Dissertation Phase-II · Jimil Jingar · PIET, Parul University

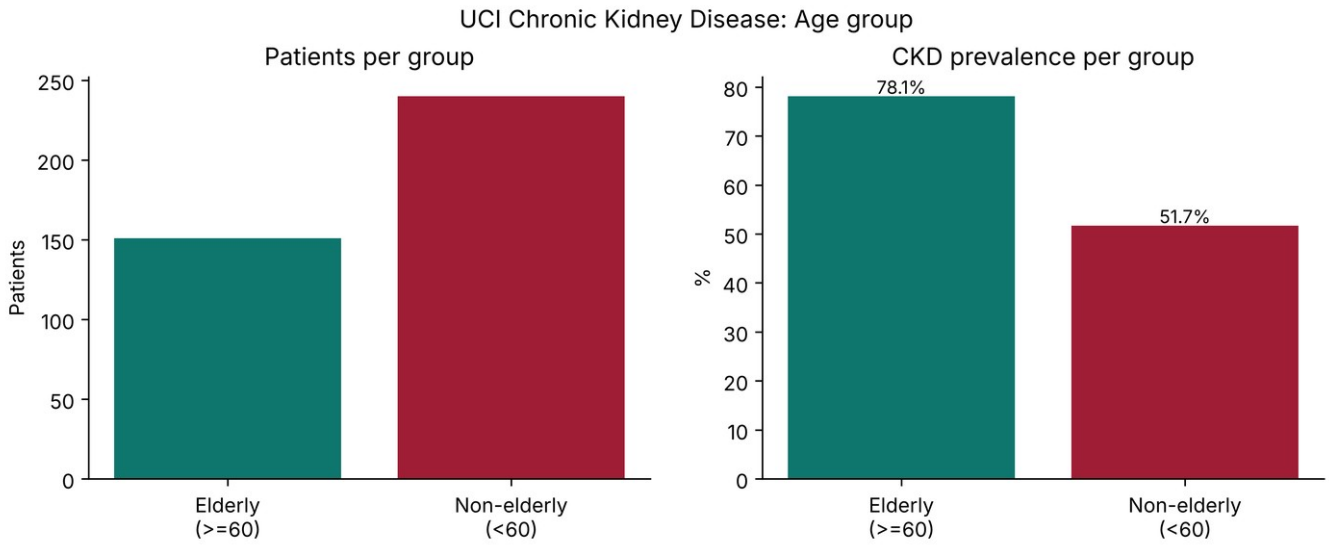
	Value
Dataset	UCI Chronic Kidney Disease
Feature set	Full clinical panel (24 features)
Sensitive attribute	Age group
Patients	391
Features	24
Groups	Elderly (>=60), Non-elderly (<60)
Decision threshold	0.500
CV protocol	stratified 5-fold, 10 repeats (baseline), 5 repeats (mitigation)

Key findings

• Best baseline model: **SVM** (accuracy 0.997, balanced accuracy 0.998, AUC 1.000).
• Bias detection: in **6 of 7** models, CKD is detected least often in the **Non-elderly (<60)** group (Equal Opportunity gap 0.007–0.046).
• Even a perfect classifier has DP diff = 0.265 and DI = 0.661 here, because CKD prevalence differs between groups; error-rate metrics (Equal Opportunity, Equalized Odds) are the clinically meaningful fairness criteria.
• Logistic Regression: best mitigation (balanced accuracy loss ≤ 3 points) = **RW + TO (hybrid)**, Equalized Odds gap 0.016 → 0.007, balanced accuracy 0.996 → 0.998.
• XGBoost: best mitigation (balanced accuracy loss ≤ 3 points) = **Unawareness**, Equalized Odds gap 0.019 → 0.019, balanced accuracy 0.989 → 0.989.
• **FairCKD-Select** (proposed, nested CV): Equalized Odds gap 0.020 → **0.012** compared with accuracy-driven model selection, balanced accuracy 0.995 → 0.993.
• Statistical significance: 0 of 10 mitigation results reduce the Equalized Odds gap significantly (paired bootstrap, $p < 0.05$).

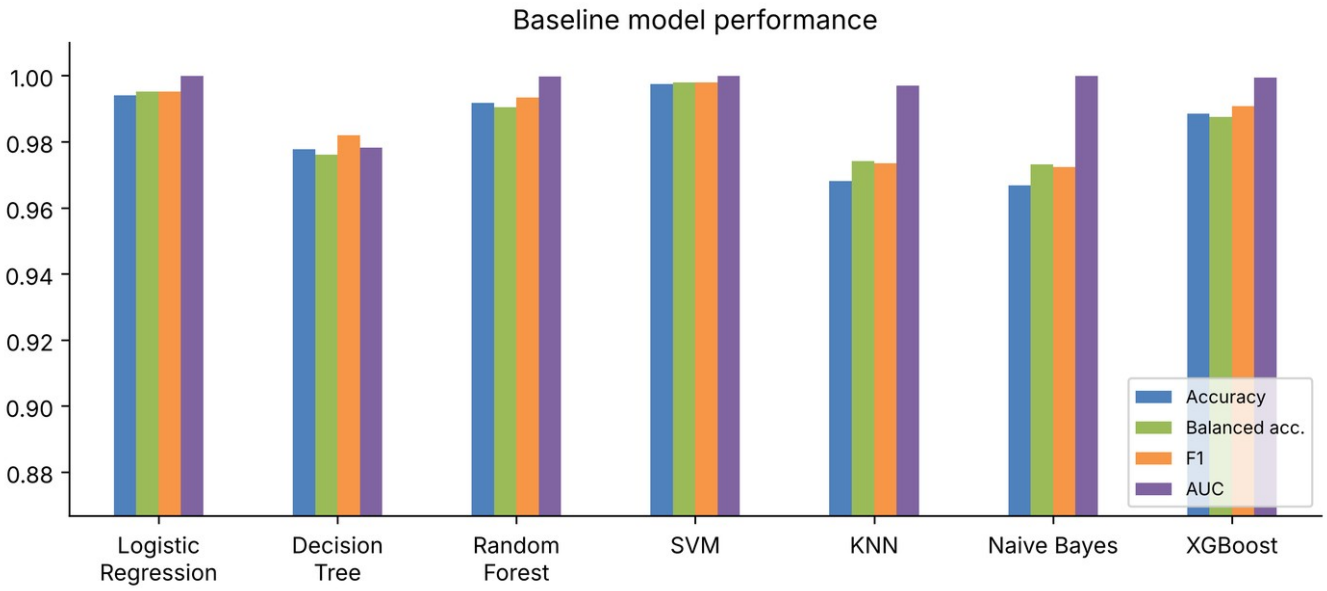
1. Dataset and groups

Group	Patients	Share (%)	CKD cases	CKD prevalence (%)	Missing values / patient
Elderly (>=60)	151	38.6	118	78.1	2.91
Non-elderly (<60)	240	61.4	124	51.7	2.17
All	391	100	242	61.9	2.46


UCI Chronic Kidney Disease: Age group
Patients per group
Patients
250
200
150
100
50
0
Elderly
(>=60)
Non-elderly
(<60)
CKD prevalence per group
%
80
70
60
50
40
30
20
10
0
78.1%
51.7%
Elderly
(>=60)
Non-elderly
(<60)

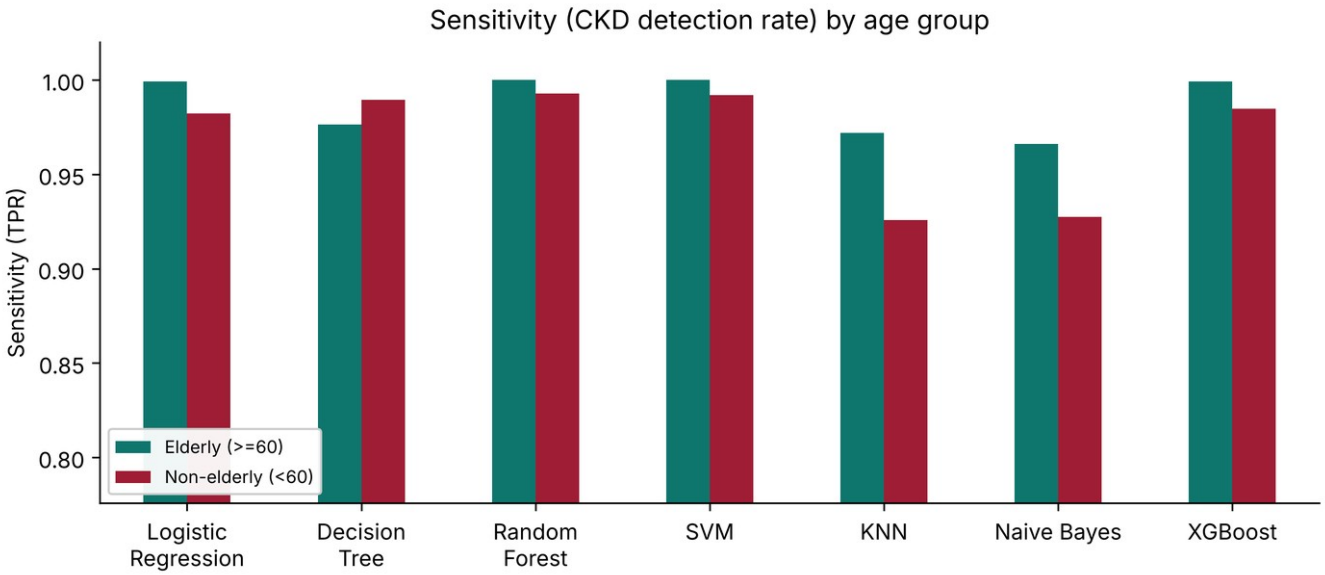
2. Baseline model performance

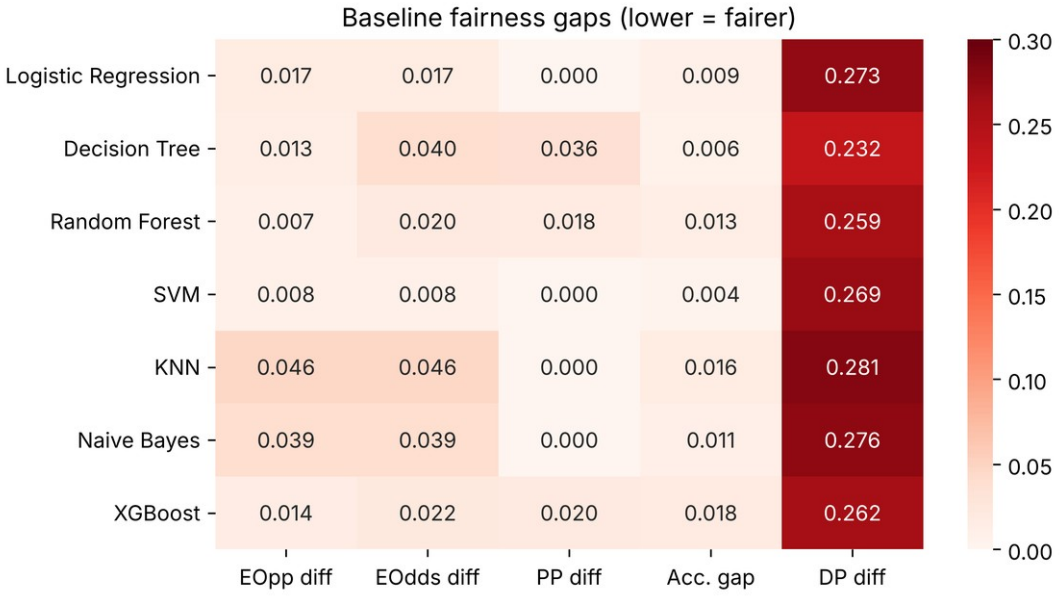
	Accuracy	Precision	Recall	F1	AUC	Balanced acc.
Logistic Regression	0.994 ± 0.004	1.000 ± 0.000	0.990 ± 0.006	0.995 ± 0.003	1.000 ± 0.000	0.995 ± 0.003
Decision Tree	0.978 ± 0.006	0.981 ± 0.006	0.983 ± 0.008	0.982 ± 0.005	0.978 ± 0.006	0.976 ± 0.006
Random Forest	0.992 ± 0.002	0.991 ± 0.002	0.996 ± 0.001	0.993 ± 0.001	1.000 ± 0.000	0.990 ± 0.002
SVM	0.997 ± 0.002	1.000 ± 0.000	0.996 ± 0.003	0.998 ± 0.001	1.000 ± 0.000	0.998 ± 0.001
KNN	0.968 ± 0.002	1.000 ± 0.000	0.948 ± 0.004	0.973 ± 0.002	0.997 ± 0.003	0.974 ± 0.002
Naive Bayes	0.967 ± 0.000	1.000 ± 0.000	0.946 ± 0.000	0.972 ± 0.000	1.000 ± 0.000	0.973 ± 0.000
XGBoost	0.988 ± 0.003	0.990 ± 0.003	0.992 ± 0.005	0.991 ± 0.002	0.999 ± 0.000	0.987 ± 0.003



3. Bias detection and fairness evaluation

	DP diff	DI ratio	EOpp diff	EOdds diff	PP diff	Acc. gap
Logistic Regression	0.273 ± 0.005	0.650 ± 0.007	0.017 ± 0.010	0.017 ± 0.010	0.000 ± 0.000	0.009 ± 0.005
Decision Tree	0.232 ± 0.008	0.695 ± 0.010	0.013 ± 0.007	0.040 ± 0.012	0.036 ± 0.011	0.006 ± 0.006
Random Forest	0.259 ± 0.002	0.669 ± 0.003	0.007 ± 0.003	0.020 ± 0.004	0.018 ± 0.004	0.013 ± 0.003
SVM	0.269 ± 0.003	0.656 ± 0.004	0.008 ± 0.005	0.008 ± 0.005	0.000 ± 0.000	0.004 ± 0.003
KNN	0.281 ± 0.007	0.630 ± 0.008	0.046 ± 0.012	0.046 ± 0.012	0.000 ± 0.000	0.016 ± 0.007
Naive Bayes	0.276 ± 0.000	0.635 ± 0.000	0.039 ± 0.000	0.039 ± 0.000	0.000 ± 0.000	0.011 ± 0.000
XGBoost	0.262 ± 0.004	0.665 ± 0.006	0.014 ± 0.005	0.022 ± 0.006	0.020 ± 0.005	0.018 ± 0.003
Perfect classifier (reference)	0.265	0.661	0.000	0.000	0.000	0.000


Sensitivity (CKD detection rate) by age group
Sensitivity (TPR)
1.00
0.95
0.90
0.85
0.80
Elderly (>=60)
Non-elderly (<60)
Logistic Regression
Decision Tree
Random Forest
SVM
KNN
Naive Bayes
XGBoost


Baseline fairness gaps (lower = fairer)
Logistic Regression 0.017 0.017 0.000 0.009 0.273
Decision Tree 0.013 0.040 0.036 0.006 0.232
Random Forest 0.007 0.020 0.018 0.013 0.259
SVM 0.008 0.008 0.000 0.004 0.269
KNN 0.046 0.046 0.000 0.016 0.281
Naive Bayes 0.039 0.039 0.000 0.011 0.276
XGBoost 0.014 0.022 0.020 0.018 0.262
EOpp diff
EOdds diff
PP diff
Acc. gap
DP diff
0.30
0.25
0.20
0.15
0.10
0.05
0.00

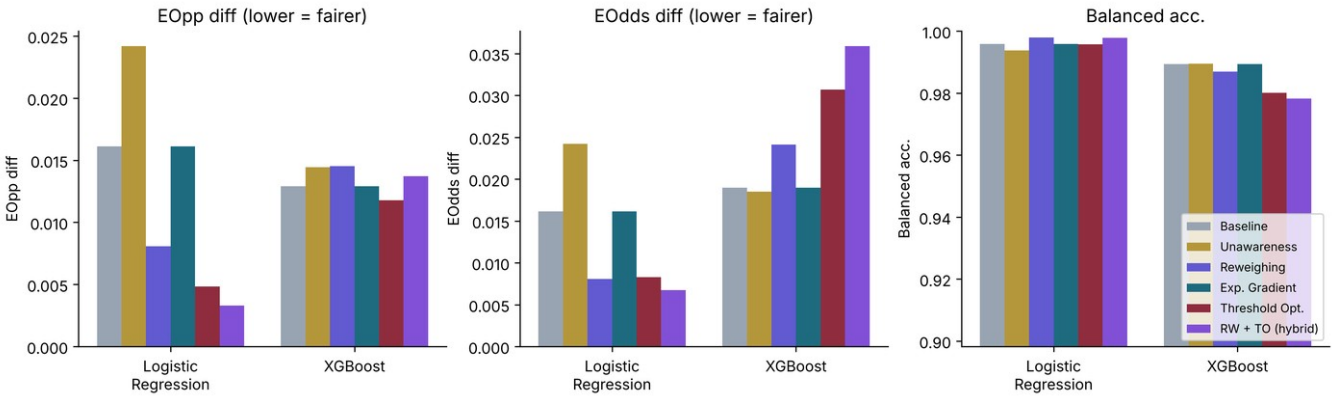
4. Bias mitigation

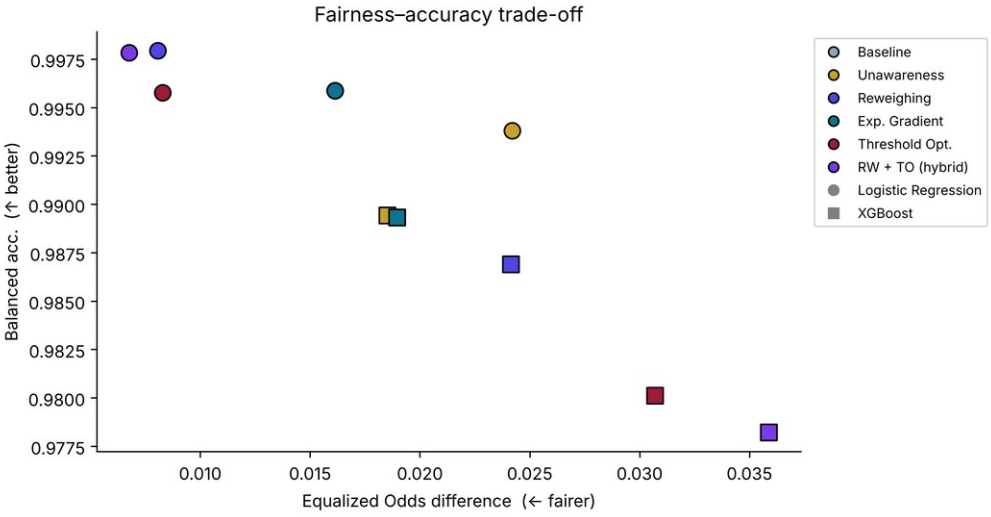
a) Performance (mean ± SD over CV repeats)

Model	Method	Accuracy	Balanced acc.	Recall	F1
Logistic Regression	Baseline	0.995 ± 0.003	0.996 ± 0.002	0.992 ± 0.004	0.996 ± 0.002
Logistic Regression	Unawareness	0.992 ± 0.002	0.994 ± 0.001	0.988 ± 0.003	0.994 ± 0.001
Logistic Regression	Reweighing	0.997 ± 0.003	0.998 ± 0.003	0.996 ± 0.005	0.998 ± 0.003
Logistic Regression	Exp. Gradient	0.995 ± 0.003	0.996 ± 0.002	0.992 ± 0.004	0.996 ± 0.002
Logistic Regression	Threshold Opt.	0.995 ± 0.002	0.996 ± 0.002	0.994 ± 0.005	0.996 ± 0.002
Logistic Regression	RW + TO (hybrid)	0.998 ± 0.002	0.998 ± 0.003	0.998 ± 0.002	0.998 ± 0.002
XGBoost	Baseline	0.990 ± 0.002	0.989 ± 0.002	0.993 ± 0.002	0.992 ± 0.002
XGBoost	Unawareness	0.990 ± 0.003	0.989 ± 0.003	0.991 ± 0.002	0.992 ± 0.002
XGBoost	Reweighing	0.988 ± 0.002	0.987 ± 0.003	0.993 ± 0.002	0.991 ± 0.002
XGBoost	Exp. Gradient	0.990 ± 0.002	0.989 ± 0.002	0.993 ± 0.002	0.992 ± 0.002
XGBoost	Threshold Opt.	0.980 ± 0.007	0.980 ± 0.006	0.978 ± 0.013	0.983 ± 0.006
XGBoost	RW + TO (hybrid)	0.977 ± 0.003	0.978 ± 0.001	0.975 ± 0.012	0.982 ± 0.002

b) Fairness gaps (lower = fairer; DI ratio: 1 = fair)

Model	Method	EOpp diff	EOdds diff	PP diff	DP diff	DI ratio	Acc. gap
Logistic Regression	Baseline	0.016 ± 0.008	0.016 ± 0.008	0.000 ± 0.000	0.273 ± 0.004	0.650 ± 0.005	0.008 ± 0.004
Logistic Regression	Unawareness	0.024 ± 0.006	0.024 ± 0.006	0.000 ± 0.000	0.277 ± 0.003	0.645 ± 0.004	0.012 ± 0.003
Logistic Regression	Reweighing	0.008 ± 0.010	0.008 ± 0.010	0.000 ± 0.000	0.269 ± 0.005	0.656 ± 0.007	0.004 ± 0.005
Logistic Regression	Exp. Gradient	0.016 ± 0.008	0.016 ± 0.008	0.000 ± 0.000	0.273 ± 0.004	0.650 ± 0.005	0.008 ± 0.004
Logistic Regression	Threshold Opt.	0.005 ± 0.004	0.008 ± 0.006	0.003 ± 0.007	0.265 ± 0.005	0.660 ± 0.007	0.004 ± 0.003
Logistic Regression	RW + TO (hybrid)	0.003 ± 0.005	0.007 ± 0.007	0.003 ± 0.007	0.263 ± 0.005	0.663 ± 0.006	0.004 ± 0.004
XGBoost	Baseline	0.013 ± 0.004	0.019 ± 0.004	0.018 ± 0.004	0.262 ± 0.002	0.664 ± 0.003	0.016 ± 0.003
XGBoost	Unawareness	0.014 ± 0.004	0.019 ± 0.004	0.015 ± 0.007	0.264 ± 0.005	0.661 ± 0.005	0.015 ± 0.004
XGBoost	Reweighing	0.015 ± 0.004	0.024 ± 0.007	0.022 ± 0.007	0.261 ± 0.004	0.666 ± 0.005	0.019 ± 0.004
XGBoost	Exp. Gradient	0.013 ± 0.004	0.019 ± 0.004	0.018 ± 0.004	0.262 ± 0.002	0.664 ± 0.003	0.016 ± 0.003
XGBoost	Threshold Opt.	0.012 ± 0.005	0.031 ± 0.019	0.012 ± 0.013	0.255 ± 0.019	0.668 ± 0.019	0.007 ± 0.008
XGBoost	RW + TO (hybrid)	0.014 ± 0.009	0.036 ± 0.028	0.013 ± 0.012	0.254 ± 0.021	0.669 ± 0.020	0.011 ± 0.010





c) Statistical significance (paired bootstrap vs. baseline, 1000 resamples; Δ = method − baseline)

Model	Method	Δ EOdds gap	95% CI (EOdds)	p (EOdds)	Δ Balanced acc.	95% CI (Bal. acc.)	p (Bal. acc.)
Logistic Regression	Unawareness	0.008	[0.000, 0.026]	0.742	-0.002	[-0.006, 0.000]	0.742
Logistic Regression	Reweighing	-0.016	[-0.042, 0.000]	0.264	0.004	[0.000, 0.011]	0.264
Logistic Regression	Exp. Gradient	0.000	[0.000, 0.000]	1	0.000	[0.000, 0.000]	1
Logistic Regression	Threshold Opt.	-0.016	[-0.042, 0.000]	0.264	0.004	[0.000, 0.011]	0.264
Logistic Regression	RW + TO (hybrid)	-0.016	[-0.042, 0.000]	0.264	0.004	[0.000, 0.011]	0.264
XGBoost	Unawareness	0.000	[0.000, 0.000]	1	0.000	[0.000, 0.000]	1
XGBoost	Reweighing	0.006	[0.000, 0.026]	1	-0.003	[-0.011, 0.000]	0.792
XGBoost	Exp. Gradient	0.000	[0.000, 0.000]	1	0.000	[0.000, 0.000]	1
XGBoost	Threshold Opt.	0.028	[-0.019, 0.117]	0.632	-0.006	[-0.021, 0.004]	0.494
XGBoost	RW + TO (hybrid)	0.052	[-0.017, 0.166]	0.372	-0.010	[-0.022, 0.000]	0.108

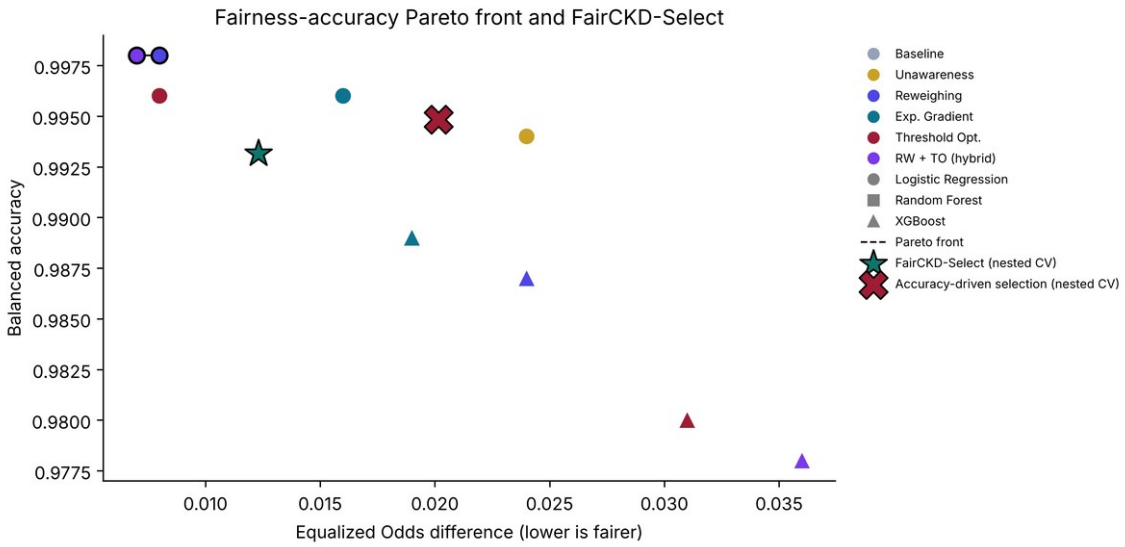
4b. FairCKD-Select (proposed method)

FairCKD-Select chooses, inside each training fold, the classifier and mitigation method with the smallest Equalized Odds gap among those whose balanced accuracy is within 3 points of the best unmitigated model (nested cross-validation). It is compared with conventional accuracy-driven selection.

Strategy	Accuracy	Balanced acc.	Recall	EOpp diff	EOdds diff	DP diff
FairCKD-Select	0.992 ± 0.004	0.993 ± 0.004	0.990 ± 0.003	0.012 ± 0.017	0.012 ± 0.017	0.268 ± 0.008
Accuracy-driven selection	0.994 ± 0.002	0.995 ± 0.001	0.990 ± 0.003	0.020 ± 0.006	0.020 ± 0.006	0.275 ± 0.003

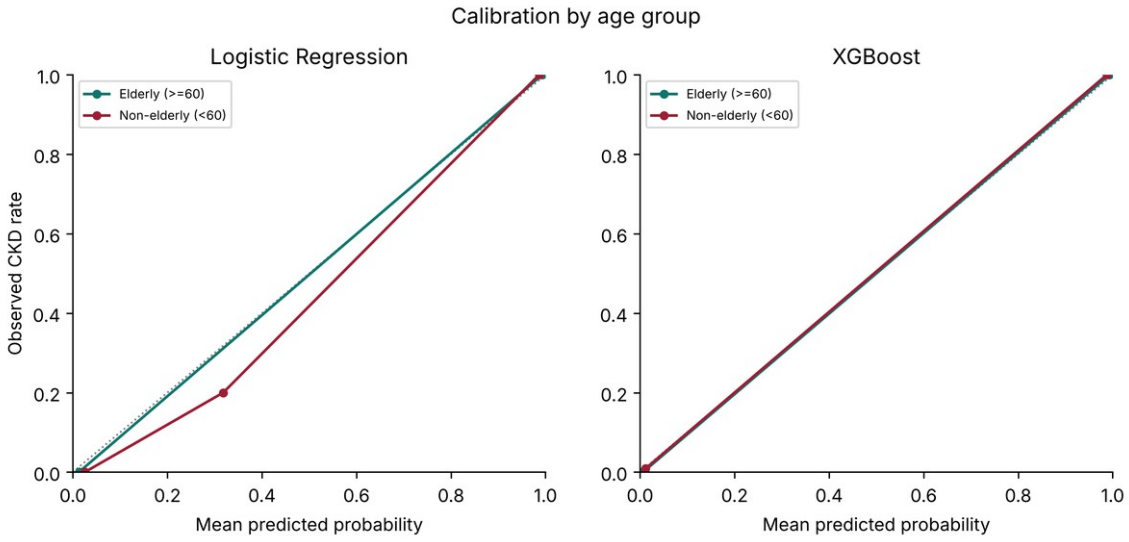
Pipelines chosen by FairCKD-Select across outer folds

Model	Method	Times selected
Logistic Regression	Baseline	2
Logistic Regression	Reweighing	2
Logistic Regression	Threshold Opt.	4
Logistic Regression	Unawareness	1
XGBoost	Threshold Opt.	1

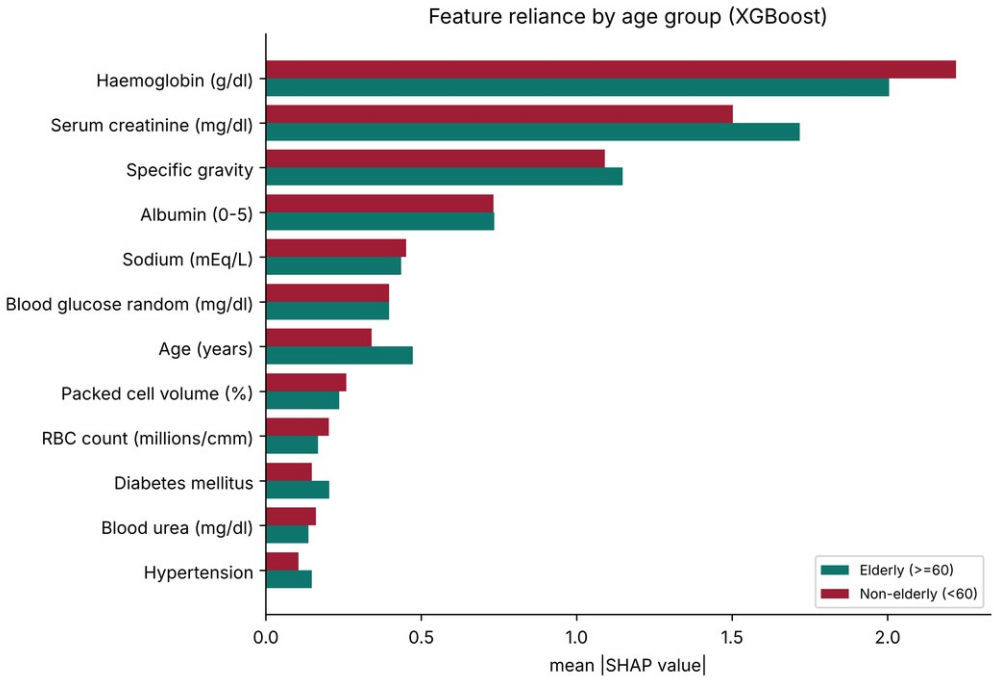


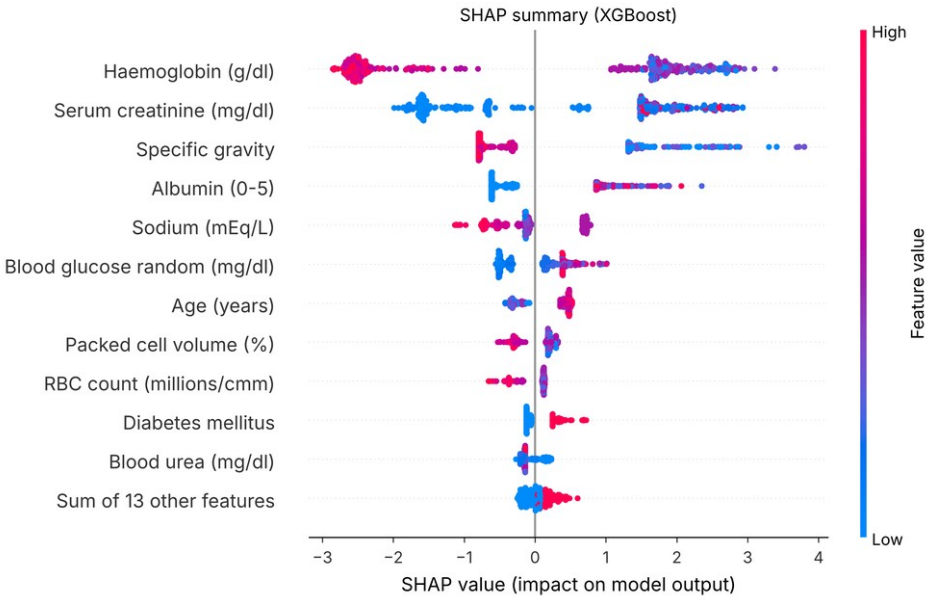
4c. Calibration by group

Group	Patients	Observed CKD (%)	Mean predicted (%)	Calibration-in-the-large (pp)	ECE	Brier	Model
Elderly (>=60)	151	78.146	77.277	-0.869	0.014	0.003	Logistic Regression
Non-elderly (<60)	240	51.667	51.471	-0.196	0.031	0.009	Logistic Regression
Elderly (>=60)	151	78.146	77.777	-0.369	0.006	0.000	XGBoost
Non-elderly (<60)	240	51.667	51.883	0.217	0.017	0.013	XGBoost


Calibration by age group
Logistic Regression
XGBoost
Elderly (>=60)
Non-elderly (<60)
Observed CKD rate
Mean predicted probability
0.0
0.2
0.4
0.6
0.8
1.0

6. Explainability (SHAP)





Protocol: repeated stratified 5-fold cross-validation (stratified on label × group); preprocessing and mitigation fitted on training folds only; mean ± SD over repeats. This system is a research prototype and not a medical device.